CMPSC 448 Midterm Project
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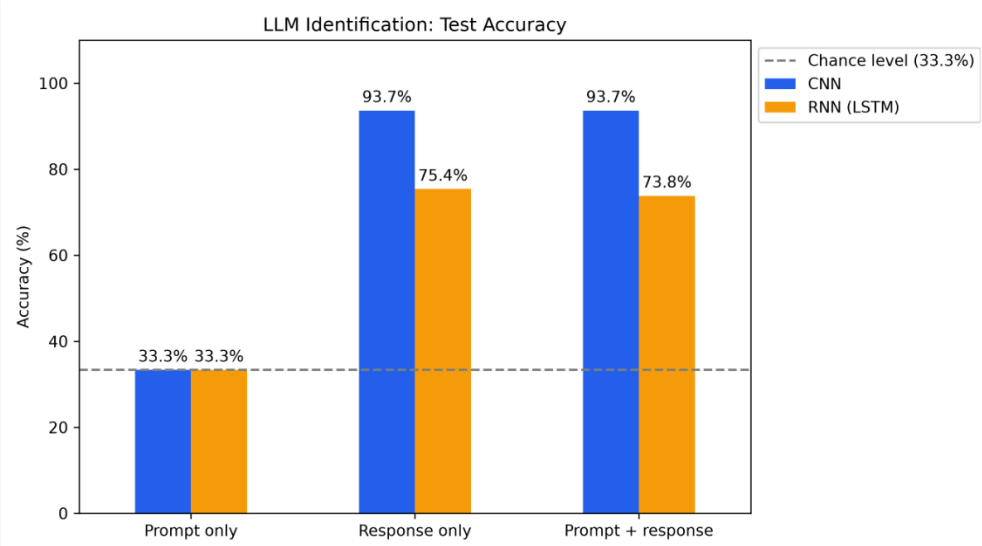
In this project we will be answering the question, “Can machine learning identify the AI behind the text?”. We test whether CNN and RNN classifiers can identify responses generated by ChatGPT, Claude, and Gemini. The models we used were ChatGPT -6.1 Sol Medium, Claude Sonnet 5.5 Medium, and Gemini Auto. We do this by answering the 4 research questions where we check whether the LLM can be identified by its response alone, we also compare prompt-only, response-only, and combined response-and-prompt inputs, we investigate generalization across domains, and distinguish linguistic differences and structural features between responses. To do this we first had to collect our data, where we created 276 unique prompts which led to a total of 828 responses. The prompts came from our interests, random questions, and past/current coursework. To try to reduce the influence from past chats in more frequently used models we put this message into the LLMs, “I am going to ask you questions for a project to gauge your response so before I do that for anything i say in this chat do not take info from other chats to influence your answers”. We split the data into 70% training, 15% validation, and 15% testing, where all responses to the same prompt stayed in the same group.

Before training we changed our text to lowercase, separated the words and punctuation into tokens, and gave each token a numerical id. Each input was limited to 512 tokens, where shorter inputs were padded to the same length. For our CNN we looked for short patterns across 3,4, or 5 tokens using 64 filters for each size. The RNN processed the text in order using one layer with 64 hidden units. Both models used 100 dimensional word embeddings and dropout of 0.5 to reduce overfitting. We trained each classifier using prompts only, responses only, and responses and prompts combined. Each model went through the training data 10 times adjusting itself after every batch of 16 examples. We used Adam to make adjustments, with a learning rate of 0.001. Cross-entropy loss measured prediction errors. After each pass through the training data, we checked performance on separate validation examples and saved the version with the lowest error score. We then measure how many predictions were correct and how well it performed across the three different LLMs.

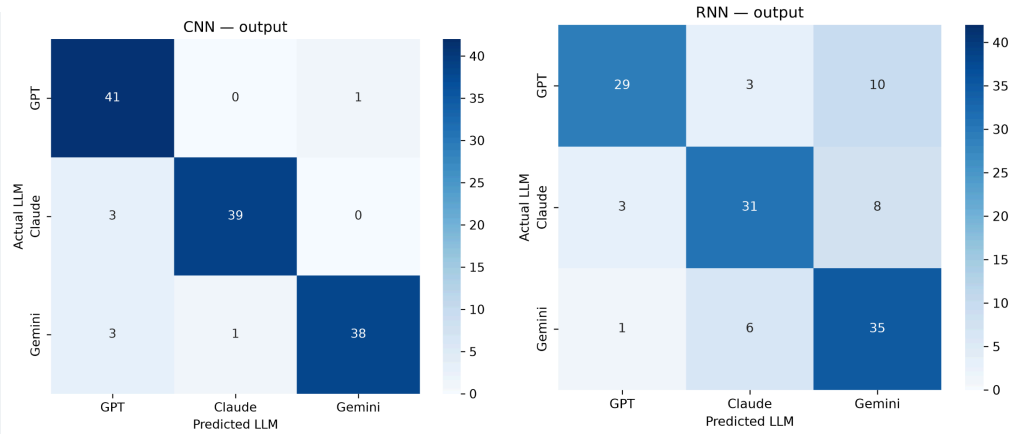
We tested all 6 models on 42 prompts for each LLM, 126 responses.

1	model	mode	saved_epoch	test_accuracy	macro_f1
2	cnn	input	8	0.3333333333333333	0.29754901960784313
3	cnn	output	7	0.9365079365079365	0.9369464772467117
4	cnn	both	10	0.9365079365079365	0.9375717673277754
5	rnn	input	9	0.3333333333333333	0.26389061700745164
6	rnn	output	6	0.753968253968254	0.7554243331907003
7	rnn	both	6	0.7380952380952381	0.737108797007175

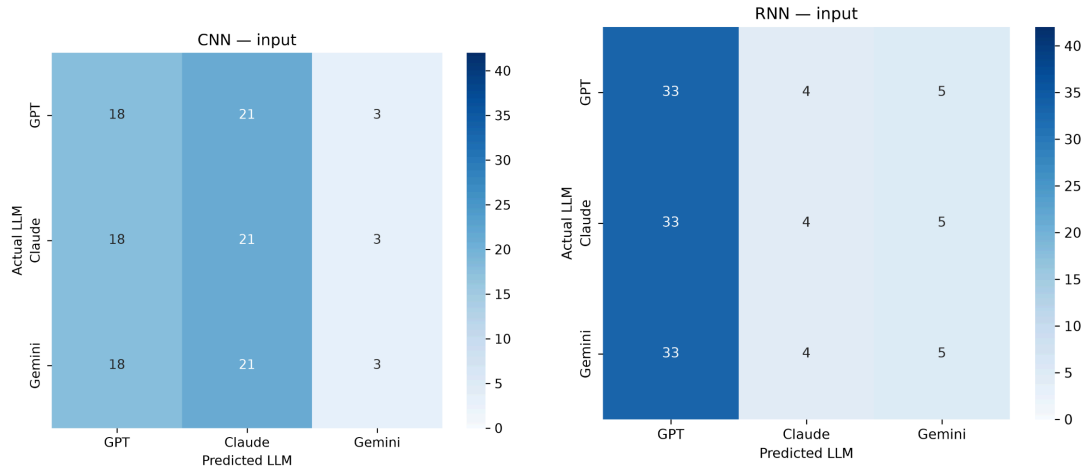
As we can see the CNN performed the best with a 93.65% accuracy, while the RNN only had an accuracy of 75.4%. Both of them performed better than the expected 33.3%, but CNN clearly performed better. We also reported the macro-F1, where the CNN scored a high of 0.937 and the RNN scored a high of 0.755.



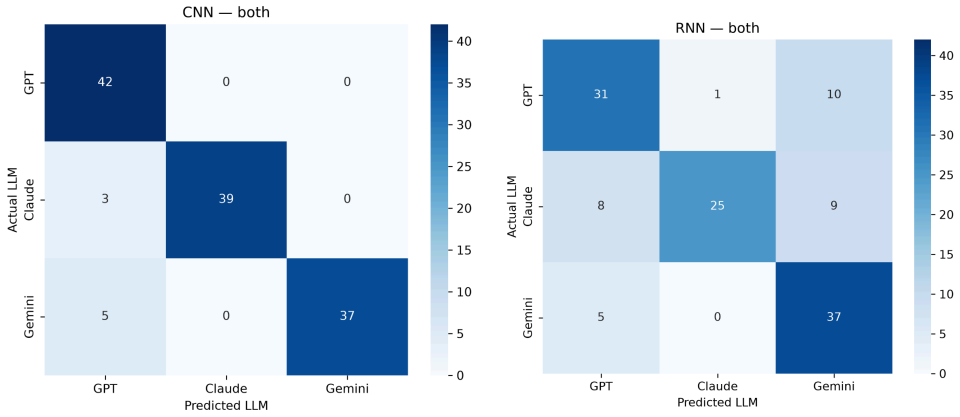
Both models were able to achieve 33.33% accuracy using the questions alone, which makes sense since all three LLM's had the same questions. Adding the question did not improve the CNN accuracy as it stayed the same, while it actually made the accuracy of the RNN go down from 75.40% to 73.81%.



In the output-only CNN model we were able to correctly identify 41/42 ChatGPT responses, 39/42 Claude responses, and 38/42 Gemini responses. This shows that CNN was able to identify ChatGPT the easiest. This was not the same in the output-only RNN model though, as it correctly identified 29/42 ChatGPT responses, 31/42 Claude responses, and 35/42 Gemini responses. We saw how RNN was constantly mistaking both models and Gemini. This shows how among both CNN and RNN Gemini was able to stay around the same amount of correctly identified responses with 35 and 38.



These are the confusion matrices from the input-only models where we are able to see how they both achieved 33.33% accuracy but in different ways, as the CNN was more split with ChatGPT and Claude getting 18 and 21 responses, while the RNN guessed ChatGPT with the majority of its responses.

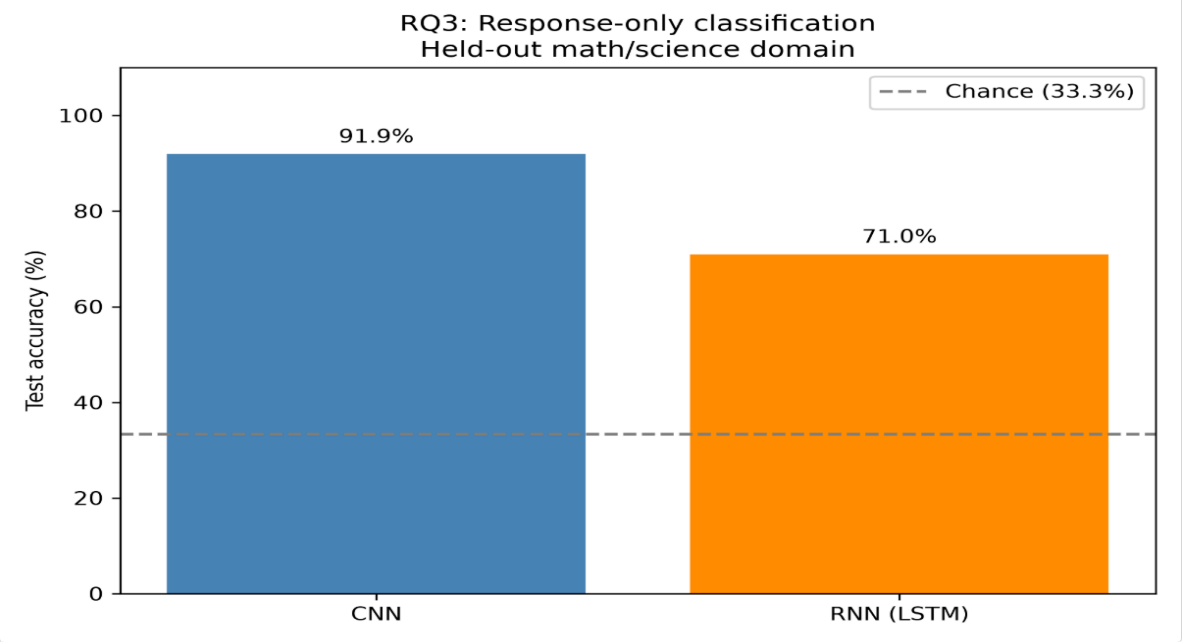


These are the confusion matrixes for the response and prompt models. Where we can see that in the CNN ChatGPT improved its correct guesses by one from the output only model, while Gemini had one fewer correct guess.. A similar story occurred in the RNN where ChatGPT and Gemini improved their correct guesses by two, but their correct guesses for Claude decreased by 6. This time the amount of Gemini correct responses was the same in CNN and RNN.

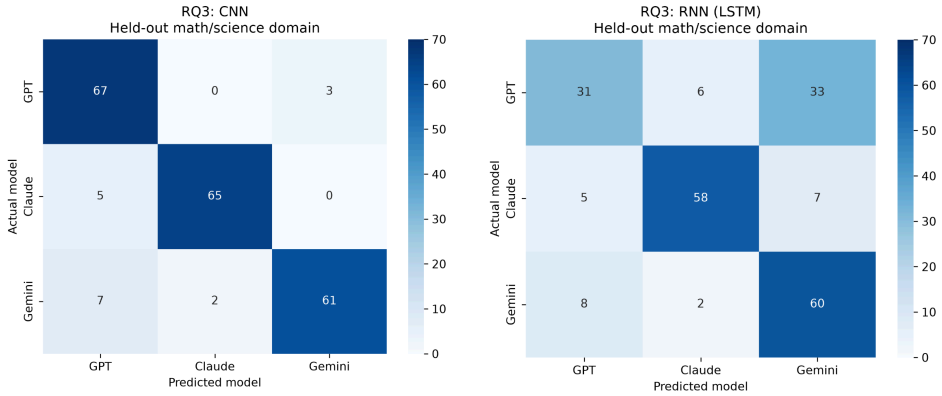
For this next part we wanted to see if the classifiers could identify responses from a category that they were not trained with. For this we altered the data and organized all prompts into specific categories like math/science, factual, opinions and more. We decided to use the math/science category for this test, this led to us using 164 prompts for training, 492 responses, 42 prompts for validation, 126 responses, and all 70 math/science prompts for testing, 210 responses. We trained new CNN and RNN models using output-only and the same training settings as before.

1	model	mode	saved_epoch	test_accuracy	macro_f1
2	cnn	output	9	0.919047619047619	0.9195605765811612
3	rnn	output	6	0.7095238095238096	0.7008943928448573

From this we can see that the CNN achieved and accuracy of 91.90%, while the RNN achieved an accuracy of 70.95%. Where the CNN had a macro-f1 of 0.920, and the RNN had a macro-f1 of 0.701.



Compared to the original output-only models CNN's accuracy decreased by 1.8%, while the RNN's accuracy also decreased by 4.4%.



The CNN was able to correctly identify 67/70 responses for ChatGPT, 65/70 for Claude, and 61/70 for Gemini. This led to ChatGPT remaining the easiest to identify LLM, but this changed in the RNN where it only correctly identified 31/70 responses mistaking 33 for Gemini. Though Gemini was able to keep its pattern of correctly guessing similar amounts of correct responses between CNN and RNN as they had a difference of 1 correct guess in the RNN. This shows how the original models could differ if they had more of a certain category in their testing data. Maybe a different category would cause the CNN to struggle, but it seems that the CNN did the best job yet again.

For our final test we wanted to see whether certain characteristics might help us distinguish the LLMs. We compared the length of responses, the format of the messages, and the diversity in vocabulary across all 828 responses. We counted the words, checked for lists, headings, bold text, and calculated the vocabulary diversity by dividing the number of unique words by the total number of words in each response.

LLM_name	Average words	Responses with bullets (%)	Responses with headings (%)	Responses with bold (%)	Average vocabulary diversity
ChatGPT	113.033	36.957	0.362	95.652	0.730
Claude	255.011	85.507	0.362	93.841	0.612
Gemini	287.312	47.101	46.014	53.261	0.628

From this test we were able to see that ChatGPT had an average of less than half the length of responses of the others. We also saw that Claude liked to respond with bullets using bullets 85.51% of the time, Where Gemini was an outlier using headings 127 times more than the other two LLMs and using bold around half as much. The average vocabulary diversity was similar for all three but ChatGPT was the outlier with a score of 0.73, this could be because of its shorter length though. From this we could see that each LLM had their own unique way of responding.

Our dataset had some limitations as the prompts came mostly from our interests and coursework. Also the categories did not contain an equal number of questions. Also each classifier only received the first 512 tokens which meant that longer responses were cut off.

For this we struggled the most with data collection, as it took us around 12 hours to manually think of and collect the data into excel. Also while collecting the data every 35-50 prompts ChatGPT stopped working for the next 2-3 hours. We learned that the CNN was more accurate than the RNN in our experiments. We also learned that ChatGPT was the easiest to identify for the CNN, while Gemini was the easiest for the RNN to identify. RQ4 confused us at first, but when we broke it down into simple measurements we were able to find differences in the LLMs. If we were to repeat this project we would try to get 500 prompts less bias in the message types. For the project Kyle Jablonsky handled the creation of the data, the planning of the project, RQ4, and the report. Mohammed Alghamadi handled the experiments for RQ1,RQ2,RQ3.